The Feedback-Star Paradigm: Redefining Social Network Clustering

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Abstract

- Enhances social network clustering.
- Incorporates user feedback to improve cluster formation.
- Overlapping cluster identification and management.
- Enhances intra-cluster cohesion and inter-cluster separability.
- Handles evolving network topology efficiently.

Objectives

- Dynamic feedback-driven adaptive clustering algorithm.
- Efficient large-scale dynamic dataset handling.
- Achieve high-quality clustering results.

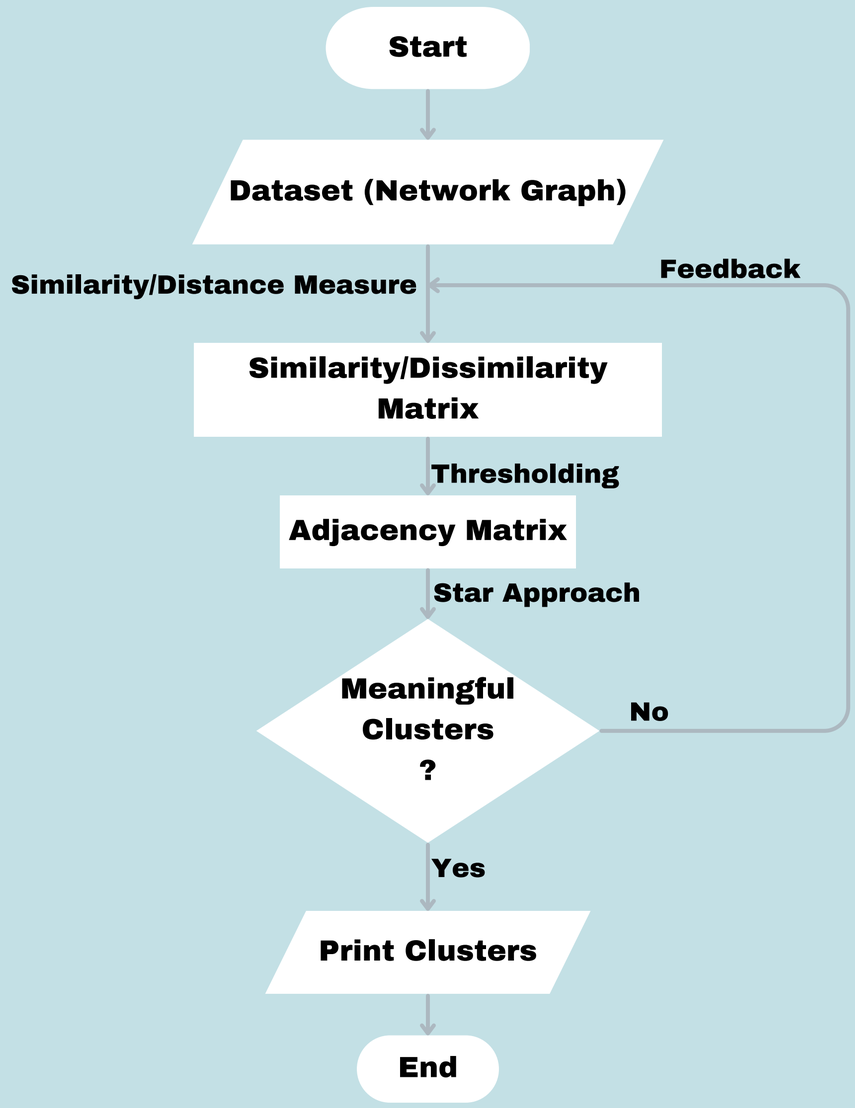
Problem Statement

- Complex, dynamic, and vital for communication and decision-making.
- Limitations of Traditional Clustering:
  - Lacks adaptability to dynamic and overlapping clusters.
  - Inefficient for large datasets.
  - The number of clusters must be defined.
- Improves clustering accuracy.

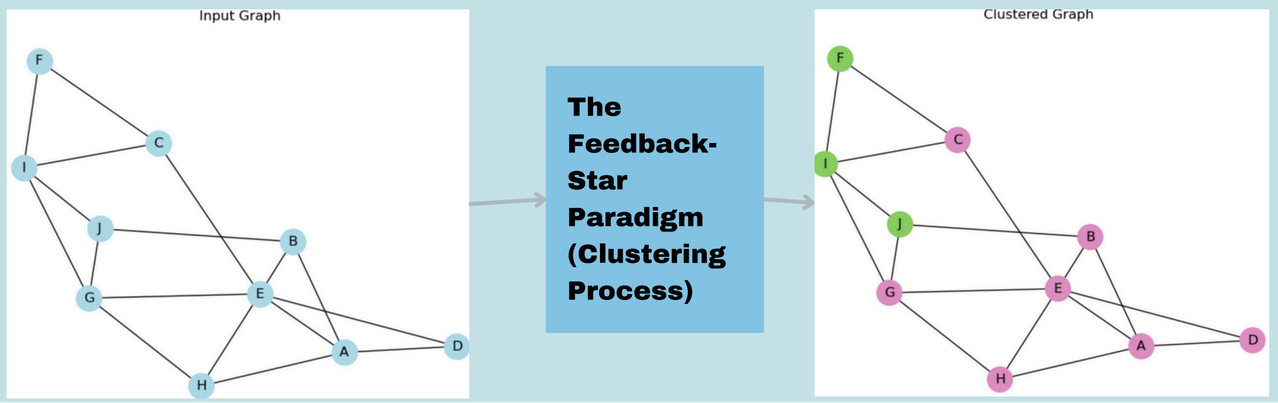
Literature Review

- Introduced star clustering
  - May introduce bias and limit scalability [1].
- Improved clustering with overlap detection and reduced order dependency
  - Lack scalability for large datasets [2].
- Hierarchical adaptation of star clustering
  - Faces challenges in real-time large-scale applications [3].

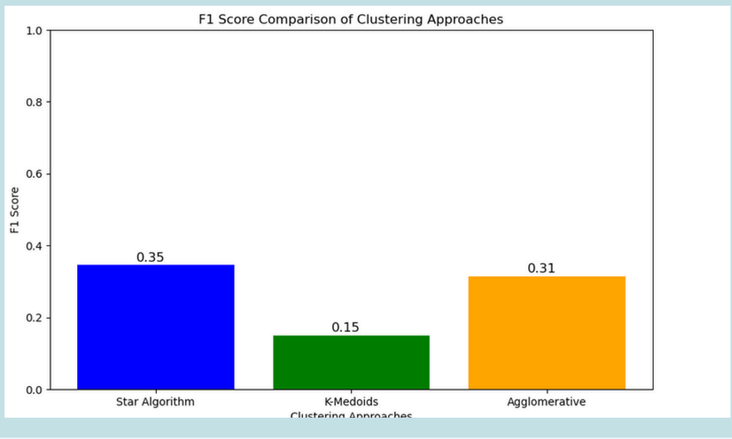
Proposed Methodology



Expected Outcome



Performance Evaluation



Conclusion

- Dynamic feedback and thresholding mechanisms.
- Improves the quality of insights drawn from social network analysis.
- Real-world applications, e.g. Survey and Decision-making Systems, Data Analysis, Market Analysis

References

[1]. J. A. Aslam, E. Pelekhov, and D. Rus, "The Star Clustering Algorithm for Static and Dynamic Information Organization," Journal of Graph Algorithms and Applications, vol. 8, no. 1, pp. 95–129, 2004.

[2]. A. P´erez Su´arez and J. E. Medina Pagola, "A Clustering Algorithm Based on Generalized Stars," Journal of Artificial Intelligence Research and Development, vol. 35, no. 2, pp. 125–145, 2023.

[3]. R. Gil-Garc´ ıa and A. Pons-Porrata, "Hierarchical Star Clustering Algorithm for Dynamic Document Collections," in Proceedings of the 13th Iberoamerican Congress on Pattern Recognition (CIARP 2008), vol. 5197, Springer-Verlag, 2008, pp. 187–194.